



### Flexible e-skin emulates natural response to stimuli like poking or scalding

Researchers at Stanford University in California have created electronic skin capable of emulating the natural response of a finger, toe, or limb to stimuli like poking or scalding. Flexible e-skin traditionally relies on rigid semi-



*The 'e-skin' is a soft, flexible sensor with integrated circuits (Credit: Jiancheng Lai and Weichen Wang of Bao Research Group at Stanford University)*

conductors and poses safety risks. The team has created a soft e-skin using a flexible polymer dielectric, enabling high-performance, stretchy transistor arrays. The sensor converts physical changes to electrical pulses. They have developed a synapse-like device for nerve-muscle signal transmission. In an experiment, touch triggers an electrical signal from the e-skin to the brain via the artificial synapse, causing a leg cramp. The e-skin benefits those with injuries or sensory disorders. Instead of brain implants, the team proposes peripheral nervous system implants for amputees. The team aims to make a wireless version. However, achieving full-finger coverage with touch, temperature, and pressure response requires further development.

### Hybrid system with remote control proposed for autonomous vehicles

Researchers at the Massachusetts Institute of Technology (MIT) have proposed a hybrid system where autonomous vehicles handle easy scenarios while remote human operators manage complex manoeuvres. The researchers addressed autonomous vehicles' merging challenges by



*Traffic rotary and peak fall foliage -New Hampshire (Credit: <https://news.mit.edu>)*

allowing remote human supervisors to take control temporarily during on-ramp merging. They set time limits for supervisors and explored the feasibility of a few supervisors managing a large fleet of vehicles. As vehicles coordinate, the need for remote supervisors decreases, enhancing reliability during merging. The team focused on merging, a difficult task for autonomous vehicles in highway driving. Automating highway driving saves time. Remote human supervisors can guide vehicles during critical moments, eliminating the need for passenger intervention.

### AI uses reflections from shiny objects to improve autonomous vehicles' visibility

MIT and Rice University researchers have developed ORCa, an AI-powered system that transforms shiny objects into functional cameras by capturing and mapping reflections. This technique improves autonomous vehicles' visibility by



*Researchers from MIT and Rice University have created a computer vision technique that leverages reflections to image the world by using them to turn glossy objects into 'cameras,' enabling a user to see the world as if they were looking through the 'lenses' of everyday objects like a ceramic coffee mug or a metallic paperweight (Credit: Courtesy of the researchers)*

utilising reflections for observing obstacles and offering potential drone imaging applications. The researchers did multiple-angle photography to capture reflections. ORCa uses machine learning to convert the object's surface into a virtual

sensor, capturing light and reflections for each virtual pixel. It models the 3D environment from the object's perspective using virtual pixels. ORCa estimates object depth and shapes using multiview reflections. It models the scene as a 5D radiance field, capturing intensity, direction, and light rays. This aids in accurate depth estimation and revealing hidden features. Users can position virtual cameras, visualise perspectives, and modify object appearance.

### AI and autonomous off-road driving tech prototypes developed for Jeep

A team led by Neda Cvijetic, Head of Artificial Intelligence (AI) & Autonomous Driving at Stellantis, tested the latest automated off-road driving technology prototypes designed exclusively for the Jeep brand in Moab, Utah. The electrified Jeep Grand Cherokee 4xe models feature advanced AI and